



FAST NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

LAB TASK: 01

PROGRAMMING FUNDAMENTAL

FIRST YEAR- 1ST SEMESTER (FALL)

SUBMITTED BY:

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TASK 1

PAC

GIVEN INPUTS	REQUIRED RESULTS
Student Roll Number Student Name Subject 1 Marks Subject 2 Marks Subject 3 Marks Subject 4 Marks Subject 5 Marks	Total Obtained Marks Percentage Grade
PROCESSING REQUIRED	SOLUTION ALTERNATIVES
$total_obtained_marks = subj_1_marks + subj_2_marks + subj_3_marks + subj_4_marks + subj_5_marks$ $percentage = total_obtained_marks * 100 / 500$ Grades Division (based on Percentage) A: ≥ 85 B: < 85 and ≥ 70 C: < 70 and ≥ 55 F: < 55	Percentage: - Use the Given Formula to find Percentage - Use total / 5 instead as simplified form Grading *Use the Percentage for Grade Assigning - Use the Total Marks for Grade Assignment

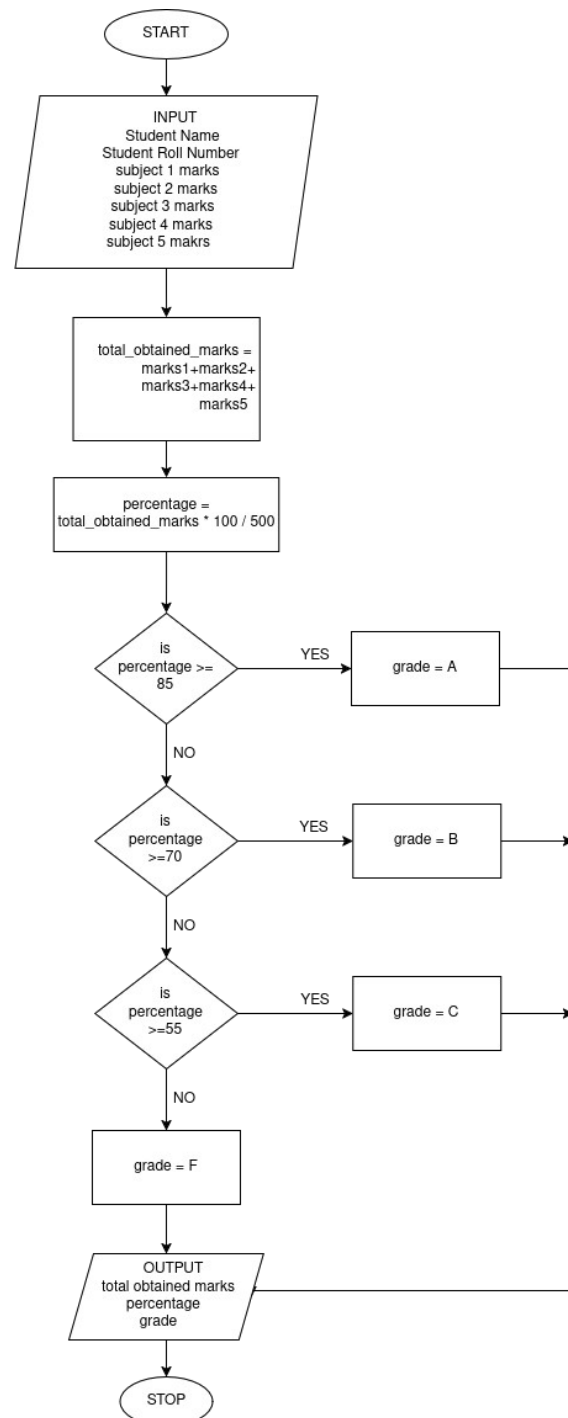
IPO

INPUT	PROCESSING	MODULE REFERENCE	OUTPUT
Student Roll Number Student Name Subject 1 Marks Subject 2 Marks Subject 3 Marks Subject 4 Marks Subject 5 Marks	Enter Student Roll Number Enter Student Name Enter Subject 1 Marks Enter Subject 2 Marks Enter Subject 3 Marks Enter Subject 4 Marks Enter Subject 5 Marks Calculate Total Marks Calculate Percentage Find Grade Output Total Marks Output Percentage Output Grade END	Read Read Read Read Read Read Read Calc Calc Calc Print Print Print	Total Obtained Marks Percentage Grade

ALGORITHM

1. Ask the user to enter their Roll Number
2. Ask the user to enter their Name
3. Ask the user to Enter Subject 1 Marks
4. Ask the user to enter Subject 2 marks
5. Ask the user to enter subject 3 marks
6. ask the user to enter subject 4 marks
7. ask the user to enter subjcet 5 marks
8. set **total_obtained_marks** to **(marks1+marks2+marks3+marks4+marks5)**
9. set **percentage** to **(total_obtained_marks * 100 / 500)**
10. set **grade** to A if **percentage** is greater then or equal to 85
11. set **grade** to B if **percentage** is less than 85 and greater than or equal to 70
12. set **grade to C** if **percentage** is less than 70 and greater than or equal to 55
13. set **grade to F** if **percentage** is less than 55
14. display **total_obtained_marks** for the user
15. display **percentage** for the user
16. display **grade for the user**

FLOWCHART



TASK 2

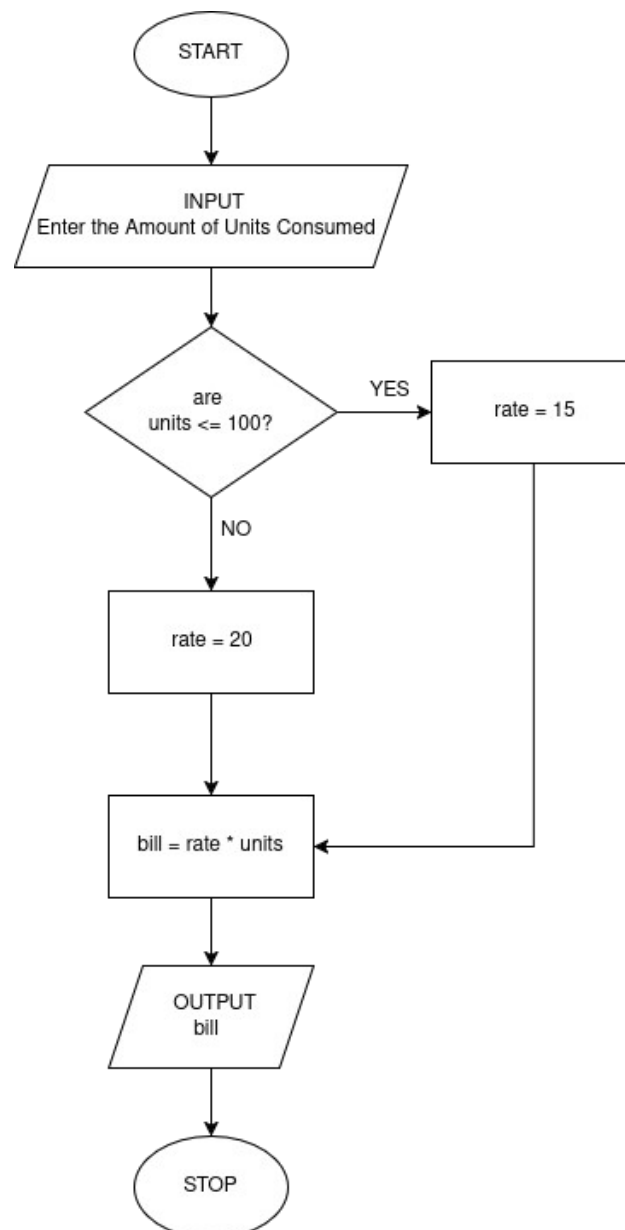
PAC

GIVEN INPUTS	REQUIRED RESULTS
units	bill
PROCESSING REQUIRED 1. Determining Rate IF <i>units</i> are less than or equal to 100 THEN <i>rate</i> = 15 ELSE <i>rate</i> = 20 2. Calculating Bill <i>bill</i> = <i>rate</i> * <i>units</i>	SOLUTION ALTERNATIVES Input Validation: 1. The Original Solution can take any Integer value 2. *We can Validate to exclude Negative Integers

ALGORITHM

1. Ask the User to enter their consumed **units**
2. set **rate** to 15 if **units** are less than or equal to 100
3. set **rate** to 20 if **units** are greater than 100
4. set **bill** = (**rate** * **units**)
5. display **bill** for the user

FLOWCHART



TASK 3

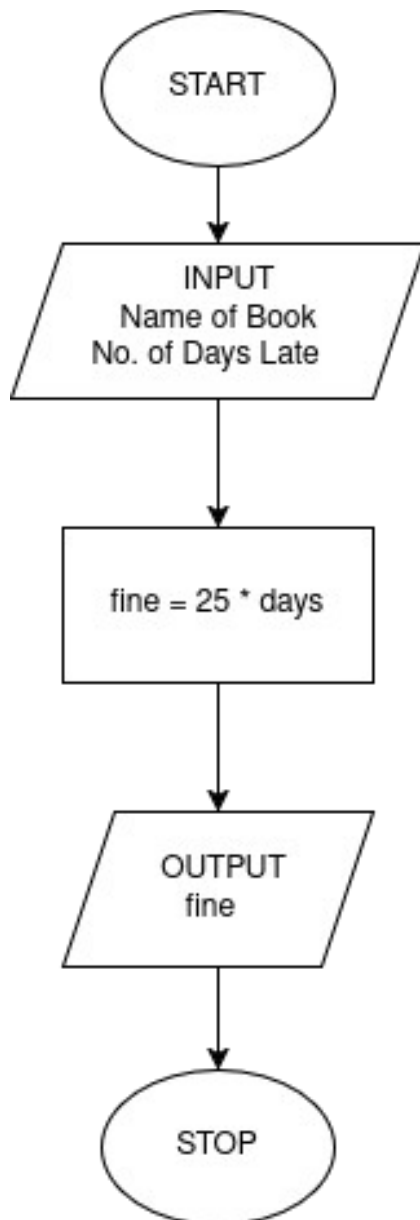
IPO CHART

INPUT	PROCESSING	MODULE REFERENCE	OUTPUT
Name of Book No. of Days Late	Enter the Book Name Enter the No. of Days that book is returned late Calculate the Fine using (fine = 25 * days) Output fine END	READ READ - CALC - PRINT	fine

ALGORITHM

1. Ask the user to enter the *Name of their Book*
2. Ask the user to enter the *number of days* the book was returned late
3. set (***fine** = 25 * days*)
4. display ***fine*** for the user

FLOWCHART



TASK 4

PAC CHART

GIVEN INPUTS	REQUIRED RESULTS
length width	shape name area
PROCESSING REQUIRED 1. Determining Shape IF <i>length</i> is equal to <i>width</i> THEN <i>shape</i> is square ELSE <i>shape</i> = rectangle 2. Calculating Area <i>area</i> = <i>length</i> * <i>width</i>	SOLUTION ALTERNATIVES Determining Shape 1. *Use the length = width logic 2. Use the Property $2 * (\text{length} + \text{width}) = 4 * \text{length}$ AND $4 * \text{width}$ Input Validation: 1. The Original Solution can take any Integer value 2. *We can Validate to exclude Negative Integers

PSEUDOCODE

START

INPUT:

ASK the User "Enter the Length"

READ Input and store in *length*

ASK the user "Enter the Width"

READ Input and Store in *width*

PROCESSING:

IF *length* is equal to *width*

THEN *shape* is square

ELSE *shape* is rectangle

SET *area* to *length* * *width*

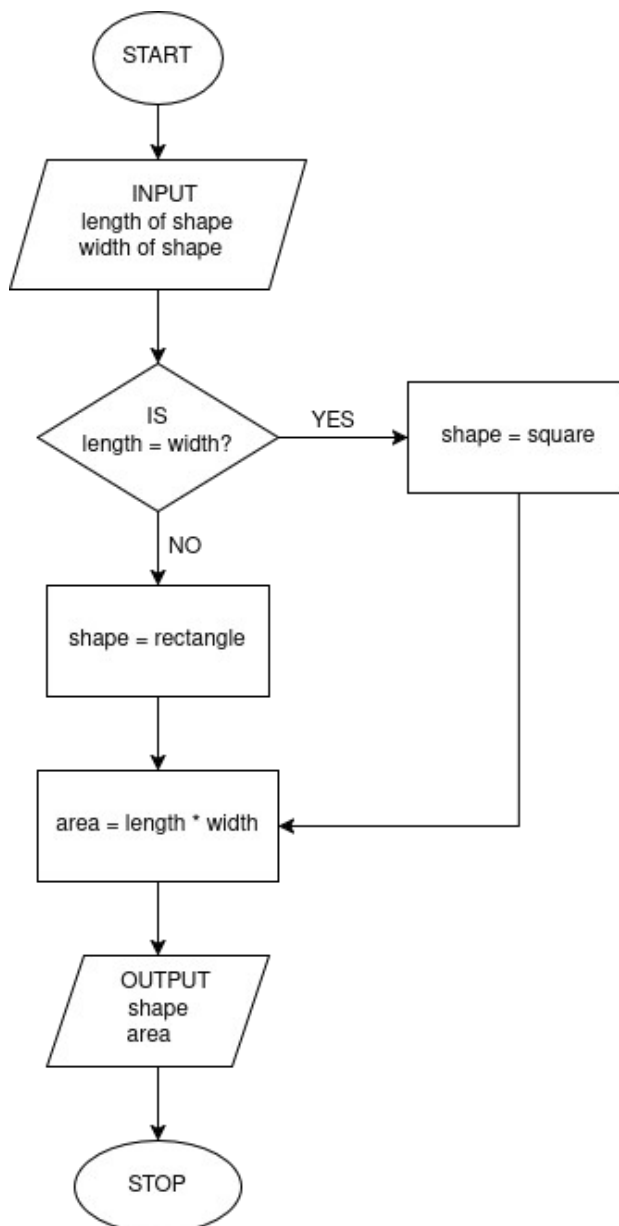
OUTPUT:

PRINT "shape is" *shape*

PRINT "area of the shape is" *area*

END

FLOWCHART



TASK 5

PAC CHART

GIVEN INPUTS	REQUIRED RESULTS
account holder's name current balance deposit amount withdraw amount	final balance
PROCESSING REQUIRED	SOLUTION ALTERNATIVES
1. Calculating Final Balance: $final_balance = current_balance + deposit_amount - withdraw_amount$	Calculating Balance *use the complete formula - find the net difference (deposit amount – withdraw amount) and then add that to current balance Input Validation: - The Original Solution can take any Integer value *We can Validate to exclude Negative Integers from deposit and withdrawal values

IPO CHART

INPUT	PROCESSING	MODULE REFERENCE	OUTPUT
Account holder's name current balance deposit amount withdrawal amount	Enter the Name of Account Holder Enter the Current Balance Enter the Deposit Amount Enter the Withdrawal Amount Calculate Final Balance Output Final Balance END	READ - READ READ READ CALC PRINT	Final Balance

ALGORITHM

1. Ask the user to Enter the *Name of Account Holder*
2. Ask the user to enter the *Current Balance*
3. Ask the user to enter the *Deposit Amount*
4. Ask the user to enter the *Withdrawal Amount*
5. Set the $final_balance = (current_balance + deposit_amount - withdraw_amount)$
6. Display the *final_balance* to the user

TASK 6

PAC CHART

GIVEN INPUTS	REQUIRED RESULTS
temperature in Celsius	temperature in Fahrenheit temperature classification
PROCESSING REQUIRED	SOLUTION ALTERNATIVES
1. Converting Temperature: $tempF = (tempC * 9 / 5) + 32$ 2. Temperature Classification IF $tempC$ is less than 15 $temp_class$ is COLD ELSEIF $tempC$ is between or equal to 15 and 30 $temp_class$ is MODERATE ELSE $tempC$ is greater than 30 $temp_class$ is HOT	Calculating Temperature in Fahrenheit 1. *use the given formula 2. use alternate formula $(tempC * 1.8) + 32$

PSEUDOCODE

START

INPUT

 ASK the user “Enter the Temperature in Celsius”

 READ the input and store in $tempC$

PROCESSING

 SET Temperature in Fahrenheit to $tempF = (tempC * 9 / 5) + 32$

 IF Temperature in Celsius is less than 15

 THEN SET $temp_class$ to COLD

 ELSE IF Temperature in Celsius is between or equal to 15 and 30

 THEN SET $temp_class$ to MODERATE

 ELSE Temperature in Celsius is greater then 30

 THEN SET $temp_class$ to HOT

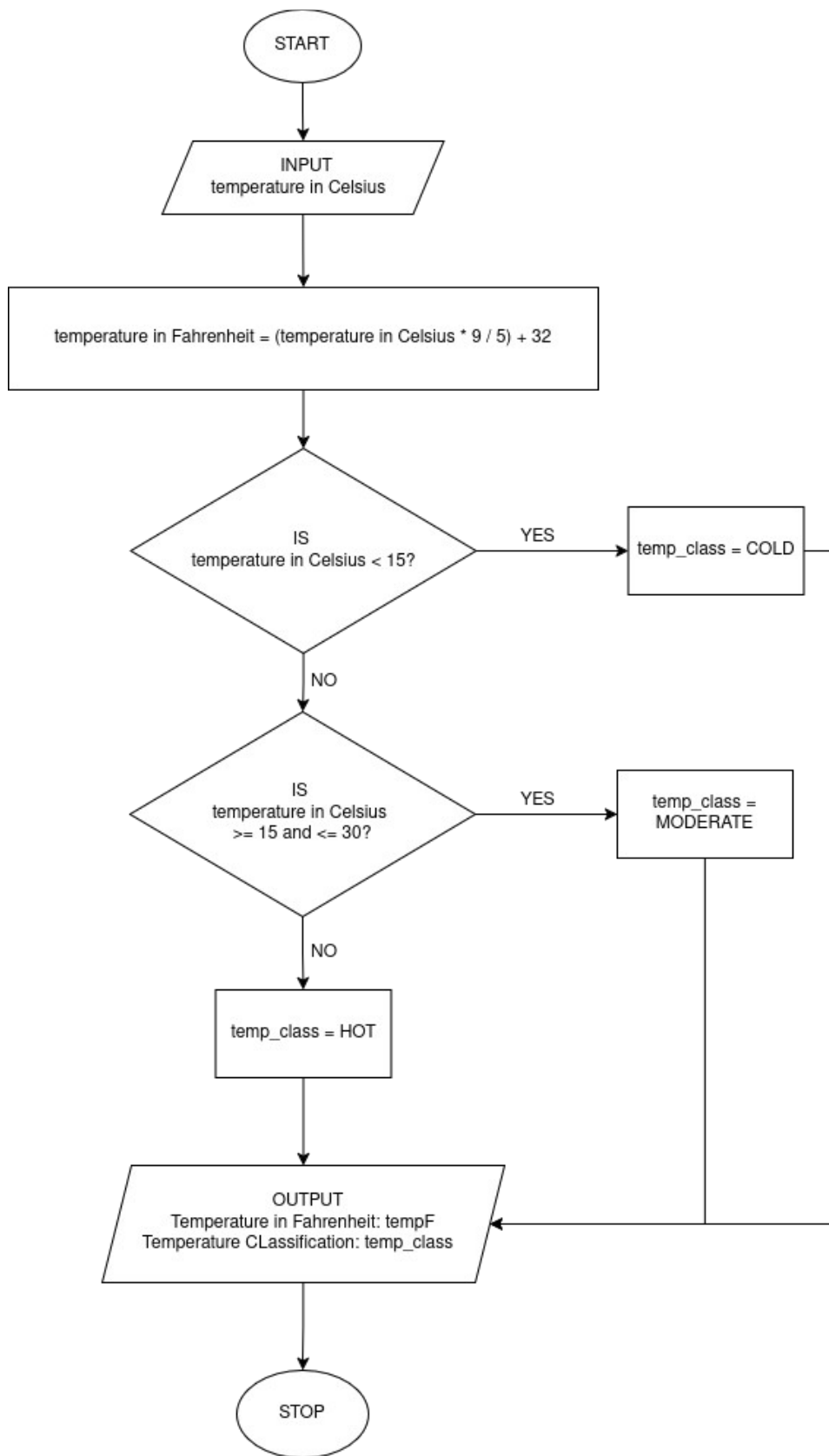
OUTPUT

 PRINT “The Temperature in Fahrenheit is” $tempF$

 PRINT “This Temperature is classified as” $temp_class$

STOP

FLOWCHART



TASK 7

PAC CHART

GIVEN INPUTS	REQUIRED RESULTS
Price of Product 1 Price of Product 2 Price of Product 3 Price of Product 4	Final AMount
PROCESSING REQUIRED	SOLUTION ALTERNATIVES
1. Total Price: $total = price1 + price2 + price3 + price4$ 2. Discount Applicable or not? IF $total$ is greater than 5000 THEN $net\ cost = price - (0.1 * price)$ ELSE $net\ cost = price$	 1. Fixed Discount for all costs above 5000 2. Ability to Set a Custom Value for Discount based on changing conditions Input Validation: 1. The Original Solution can take any Integer value 2. *We can Validate to exclude Negative Integers from prices

IPO CHART

INPUT	PROCESSING	MODULE REFERENCE	OUTPUT
Price of Product 1 Price of Product 2 Price of Product 3 Price of Product 4	Enter the Price of product 1 Enter the Price of product 2 Enter the Price of product 3 Enter the Price of product 4 Calculate Total Price IF total price is greater than 5000 $net\ price = total\ price - (0.1 * total\ price)$ ELSE $net\ price = total\ price$ output net price(Final Amount) END	READ READ READ READ CALC DECISION CALC CALC PRINT	Final Amount

ALGORITHM

1. Ask the user to enter the price of product 1: price1
2. Ask the user to enter the price of product 2: price2
3. Ask the user to enter the price of product 3: price3
4. Ask the user to enter the price of product 4: price4
5. Set $total_price = price1 + price2 + price3 + price4$
6. IF total price is greater than 5000,
7. THEN set $net_price = total_price - (0.1 * total_price)$
8. ELSE set $net_price = total_price$
9. Output Final Amount for the user: net_price

TASK 8

PAC CHART

GIVEN INPUTS	REQUIRED RESULTS
Applicant Name Applicant Age	Eligibility to Drive
PROCESSING REQUIRED	SOLUTION ALTERNATIVES
1. Eligible or Not? IF <i>age</i> is greater than or equal to 18 THEN <i>eligibility</i> = TRUE ELSE <i>eligibility</i> = FALSE	1. Use Constant Standard Age at 18 2. Option to choose between 18 and 21

PSEUDOCODE

START

INPUT

ASK the user "Enter the Name of Applicant"

READ the input and store in *name*

ASK the user "Enter the age of Applicant"

READ the input and store in *age*

PROCESSING

IF *age* >= 18

THEN *eligibility* = TRUE

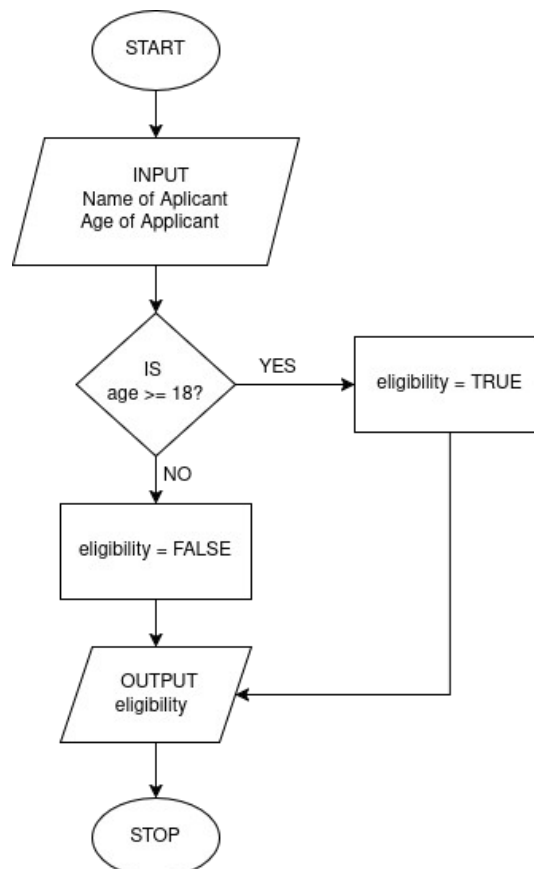
ELSE *eligibility* = FALSE

OUTPUT

PRINT "The Applicant" *name* "'s eligibilty for Driving is" *eligibility*

STOP

FLOWCHART

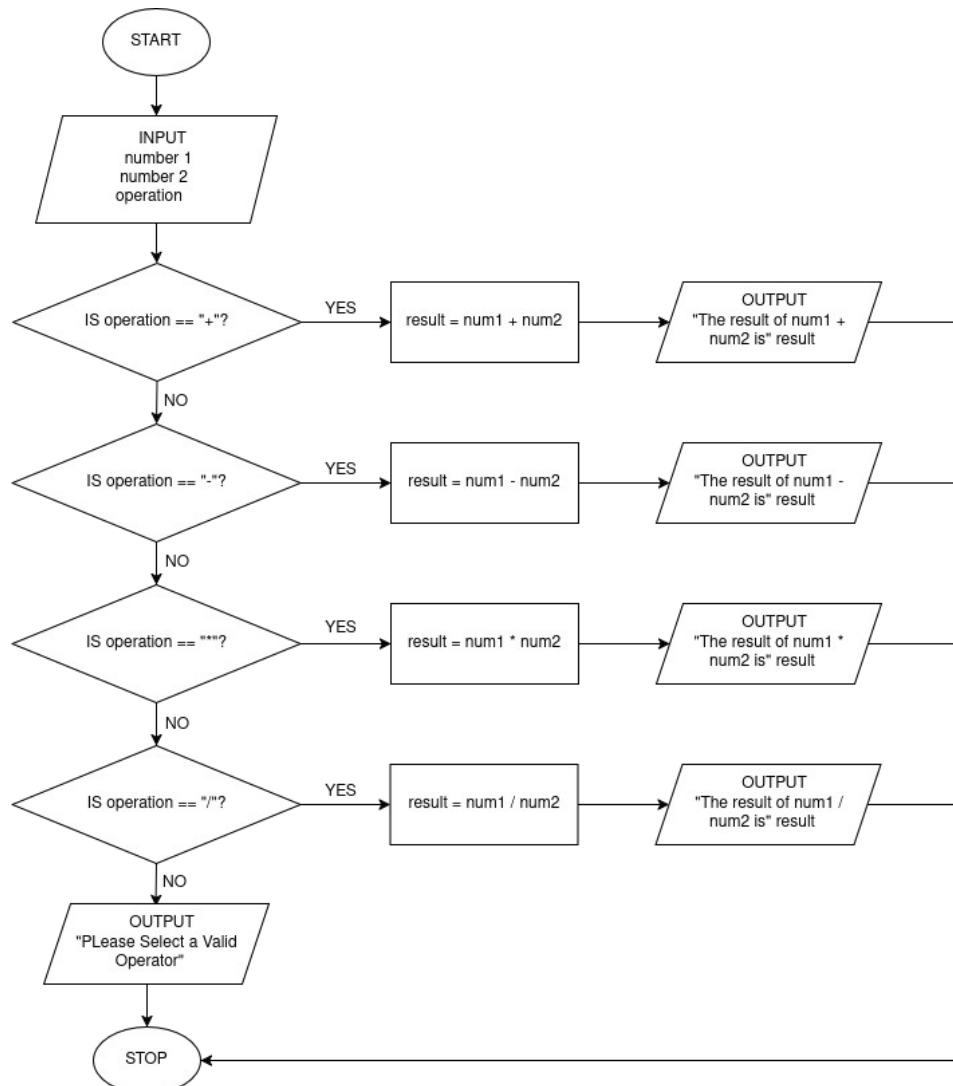


TASK 9

ALGORITHM

1. START
2. Ask the user to enter number 1; *num1*
3. Ask the user to enter number 2; *num2*
4. Ask the User to Enter 1 option from "+", "-", "*", and "/"; *operation*
5. IF *operation* is "+"
6. THEN $result = num1 + num2$
7. PRINT for the user "The result of $num1 + num2$ is"; *result*
8. ELSE IF *operation* is "-"
9. THEN $result = num1 - num2$
10. PRINT for the user "The result of $num1 - num2$ is"; *result*
11. ELSE IF *operation* is "*"
12. THEN $result = num1 * num2$
13. PRINT for the user "The result of $num1 * num2$ is"; *result*
14. ELSE IF *operation* is "/"
15. THEN $result = num1 / num2$
16. PRINT for the user "The result of $num1 / num2$ is"; *result*
17. ELSE PRINT for the user "please enter a valid operator"
18. END

FLOWCHART



SCRATCH

Aug 24 9:45 PM
Scratch 3.30.5

Code Costumes Sounds

when green flag clicked

ask "What's the first Number?" and wait

set num1 to answer

ask "What's the Second Number?" and wait

set num2 to answer

ask "Choose one of the four operations (+, -, *, /)" and wait

set operation to answer

if operation = + then

set result to num1 + num2

say join "The result of num1 + num2 is" result for 3 seconds

else if operation = - then

set result to num1 - num2

say join "The result of num1 - num2 is" result for 3 seconds

else if operation = * then

set result to num1 * num2

say join "The result of num1 * num2 is" result for 3 seconds

else if operation = / then

set result to num1 / num2

say join "The result of num1 / num2 is" result for 3 seconds

else

say "Please Enter a Valid Operator" for 3 seconds

end

when I start as a clone

create clone of myself

delete this clone

Sensing

touching mouse-pointer

touching color

color is touching

distance to mouse-pointer

ask "What's your name?" and wait

num1 1

num2 7

operation +

result 28

Sprite Sprite1

Show

Size 100

Direction 90

Sprite1

Stage

Backdrops 1

TASK 10

PAC CHART

GIVEN INPUTS	REQUIRED RESULTS
Student Name Room Rent Mess Charges Laundry Charges	Final Fee
PROCESSING REQUIRED	SOLUTION ALTERNATIVES
1. Total Fee $total\ fee = room\ rent + mess\ charges + laundry\ charges$ 2. Final Fee IF total fee is greater than 30000 THEN $final\ fee = total\ fee - (0.05 * total\ fee)$ ELSE $final\ fee = total\ fee$	Variable Discount 1. Fixed Discount for all costs above 30000 2. Ability to Set a Custom Value for Discount Input Validation: 1. The Original Solution can take any Integer value 2. *We can Validate to exclude Negative Integers from the fees

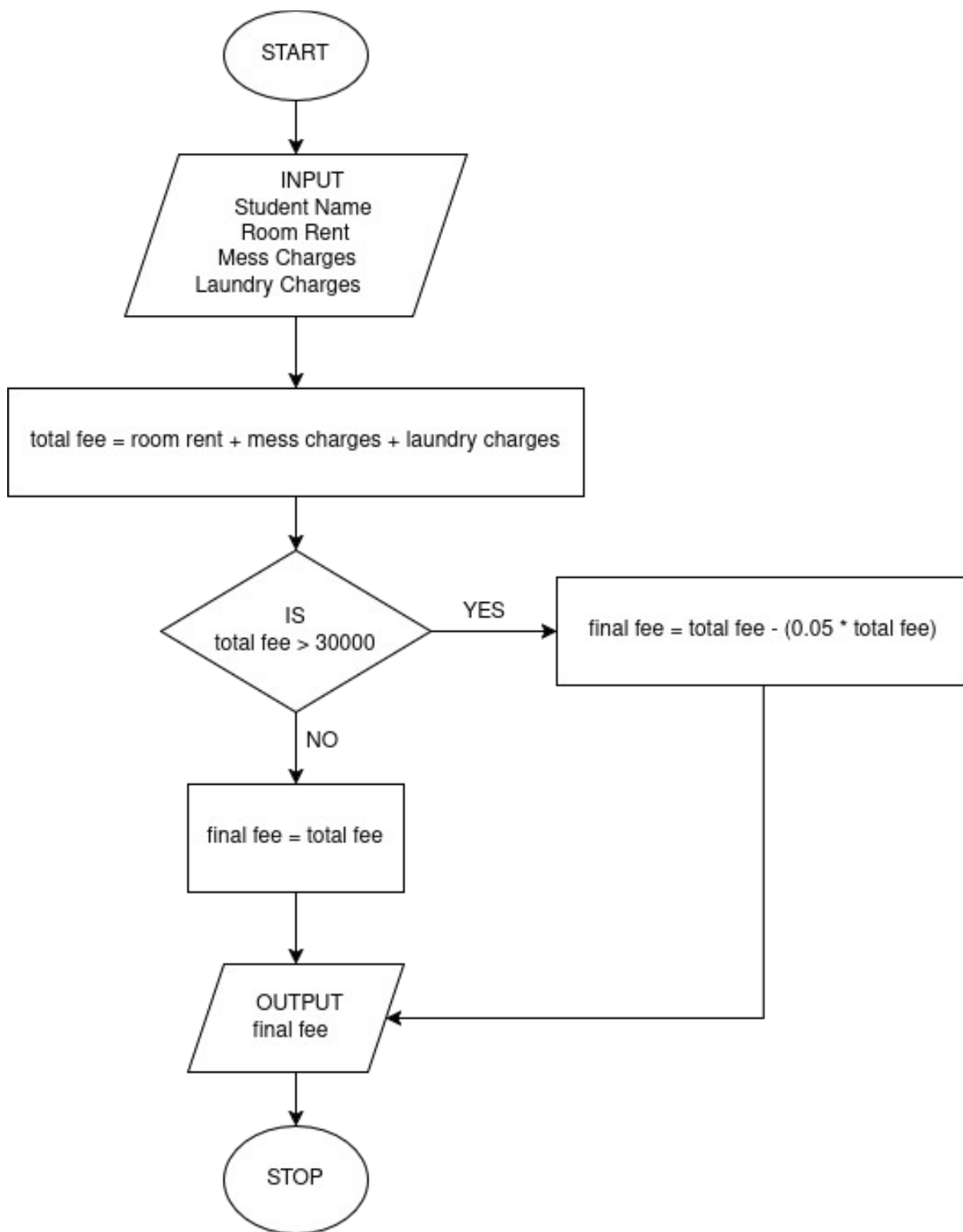
IPO CHART

INPUT	PROCESSING	MODULE REFERENCE	OUTPUT
Student Name Room Rent Mess Charges Laundry Charges	Enter the Name of Applicant Enter the Room Rent Enter the Mess Charges Enter the Laundry Charges Calculate Total Fee IF total Fee is greater than 30000 $final\ fee = total\ fee - (0.05 * total\ fee)$ ELSE $final\ fee = total\ fee$ output final fee END	READ READ READ READ CALC DECISION CALC CALC PRINT	Final Fee

ALGORITHM

1. START
2. Ask the User to Enter the Name of Applicant
3. Ask the User to Enter the Room Rent
4. Ask the User to Enter the Mess Charges
5. Ask the User to Enter the Laundry Charges
6. Set $total\ fee = room\ rent + mess\ charges + laundry\ charges$
7. IF $total\ fee$ is greater than 30000
8. THEN $final\ fee = total\ fee - (0.05 * total\ fee)$
9. ELSE $final\ fee = total\ fee$
10. Output the $final\ fee$ for the user

FLOWCHART



Scratch

Aug 24 10:19 PM

Scratch 3.30.5

Settings

File

Edit

Scratch Project

Tutorials

Code

Costumes

Sounds

username

Operators

Sensing

pick random 1 to 10

Operators

Variables

My Blocks

Variables

when clicked

ask What is the Name of Student? and wait

set name to answer

ask Enter the Room Rent and wait

set room rent to answer

ask Enter the Mess Charges and wait

set mess charges to answer

ask Enter the Laundry Charges and wait

set laundry charges to answer

set total fee to room rent + mess charges + laundry charges

if total fee > 30000 then

set final fee to total fee * 0.05 + total fee

else

set final fee to total fee

say join The Final Fee for you is final fee for 3 seconds

Dee: name

room rent

Dee: mess charges

total fee

final fee

Dee: laundry charges

The Final Fee for you is 31350

Sprite

Dee

Size

Direction

Stage

Backdrops